



Direct Impacts of Water Management on Cucumber (*Cucumis sativus* L.) Yield Components and Performance

*¹Donatus O. Igbojionu, ¹Valentine N. Ekejiuba, ²Juliet Nnennaya. Igbojionu, ² Cabrine Ifeoma. Chiemeka, ³Patience C. Ezema, ³Helen Sampson

¹ Department of Agricultural & Bio-environmental Engineering Technology, Federal College of Land Resources Technology, Owerri, Nigeria.

²Department of Soil Science and Technology, Federal College of Land Resources Technology, Owerri, Nigeria.

³Department of Crop Production Technology, Federal College of Land Resources Technology, Owerri, Nigeria.

*Corresponding authors' email: igbojionudonatus@gmail.com

ABSTRACT

This study, conducted at the Students' Demonstration Farm from September to November 2023, evaluated the impact of irrigation strategies on cucumber yield and performance. The results demonstrated that irrigating at 100% of crop evapotranspiration (ET_c, treatment W1) with a daily interval produced the peak yield of 121 ton/ha, alongside the highest fruit weight (0.565 g), length (35.5 cm), diameter (5.95 cm) and number of fruits per plant (13). In stark contrast, a severe deficit of 25% ET_c (W4) caused drastic declines, which reduced yield by 91% (10.83 ton/ha), fruit weight by 68%, fruit length by 18%, fruit diameter by 14% and fruit number by 69%. Irrigation frequency significantly modulated water stress. A 10-day interval (D10) under 100% ET_c reduced yield by 40% compared to daily irrigation, highlighting the cost of infrequent watering even when seasonal water volume is adequate. A significant interaction between irrigation amount and interval revealed critical trade-offs: high-volume, infrequent irrigation (W1D10) consumed 207.5 mm of water for a 40% (72.33 ton/ha) lower yield than W1DI (121 ton/ha), making it highly inefficient. Conversely, a moderate deficit with daily irrigation (75% ET_c, W2D1) conserved 25% of water (155.6 mm) while maintaining a respectable yield of 73.5 ton/ha (61% of maximum). Low irrigation amounts (≤50% ET_c) induced chronic stress that negated any benefits of irrigation frequency. The study provides valuable insights for farmers on optimizing irrigation practices to enhance cucumber yield and address climate change challenges.

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INTRODUCTION

Cucumber (*Cucumis sativus* L.), a globally significant vegetable crop originating in the tropics, demands substantial water for optimal growth, development, and yield (Kumar et al., 2020). Renowned for its economic and health benefits, it is widely cultivated across diverse environments, including rain-fed systems, protected structures (greenhouses, polyhouses), and conventional drip irrigation in open fields. Within West Africa, Nigeria stands as a major producer, leveraging its favourable climate and soils for cucumber cultivation (FAO, 2020; Ojo et al., 2017). Despite this potential, Nigerian production faces significant constraints, including limited irrigation access (Oladele et al., 2020), high post-harvest losses (Adebayo et al., 2017), and pest and disease pressures (Ogunbi et al., 2017). Consequently, enhancing productivity and sustainability remains a critical focus for researchers and policymakers (Adeoye et al., 2020). Compounding these local challenges, water scarcity—driven by climate change and competing demands—poses a critical global threat to cucumber production (FAO, 2023). This scarcity underscores the fundamental importance of precise irrigation management, which is not merely a conservation measure but a decisive factor in optimizing yield and quality. Specifically, water availability directly governs key yield components: fruit weight, length, diameter, and number per plant (Wang et al., 2024; Hassan et al., 2024). Suboptimal irrigation, whether deficit or excess, disrupts vital

physiological processes, impairing these critical determinants of marketable output.

The timing and severity of water stress are particularly crucial. Stress during sensitive stages like flowering and fruit set drastically reduces fruit number by compromising flower development, pollen viability, and set efficiency (Hassan et al., 2024; Li et al., 2023). Subsequently, fruit development itself is highly vulnerable; fruit size (weight, length, diameter) depends on cell expansion and assimilate partitioning, processes acutely sensitive to plant water status (Silva et al., 2024). Even moderate deficits reduce cell turgor and carbohydrate transport, leading to smaller, lighter, or misshaped fruit, while severe stress can cause yield losses of 30-50% primarily through these mechanisms (Wang et al., 2024; Li et al., 2023). Conversely, over-irrigation can impair root function and nutrient uptake, reducing fruit weight and number, and increasing disease susceptibility (Martínez-Gimeno et al., 2023).

To address these challenges, advanced irrigation strategies offer promise. Techniques like Regulated Deficit Irrigation (RDI) and Partial Root-zone Drying (PRD) aim to apply controlled stress during less sensitive periods, conserving water while minimizing damage to fruit size and weight during critical development (Silva et al., 2024). Furthermore, sensor-based scheduling (using soil moisture probes, plant sensors, or ET models) allows real-time application aligned with crop demand, significantly improving water productivity by optimizing conditions for fruit number and size compared

to calendar-based methods (Zhao et al., 2025; Pereira et al., 2024).

However, defining optimal irrigation thresholds to maximize these specific yield components remains complex, as they interact with cultivar, soil, climate, and production system (open-field vs. protected) (Martínez-Gimeno et al., 2023). Therefore, a critical need exists to synthesize physiological insights and empirical data into actionable strategies targeting these crucial yield determinants under water constraints.

This study directly addresses this need by evaluating the impacts of water management strategies on cucumber yield and its key parameters. The aim is to provide practical insights for optimizing irrigation to enhance both water use efficiency and yield performance.

MATERIALS AND METHODS

Description of Study Area

The study was conducted between September and November, 2023 at the Students Demonstration and Research Farm, Federal College of Land Resources Technology, Owerri, Southern Nigeria. The Research Farm lies between Latitude $5^{\circ}24'N$ and Longitude $6^{\circ}54'E$ and occupies an elevation of 60m above mean sea level.

It also located in a rainforest zone with an annual rainfall ranges from 2000 – 2500mm and an average temperature that ranges from $27^{\circ} - 31^{\circ}C$. The relative humidity varies from 80 – 85% during the rainy season which runs from April – October and 63 – 70% during the dry season spanning November to March. The summary of the metrological data for the period of the experiment as average for 20 years is presented in Table 1. The soil was found to be predominantly sandy clay loam with dry bulk density of 1.69 g/cm³ at 15 cm, 1.53 g/cm³ at 30 cm and 1.78 g/cm³ at 45 cm depths of sampling. The bulk result agrees with the findings of Uzoho et al. (2007). The bulk densities are generally lower than the critical limit for root penetration (1.75 – 1.85 g/cm³ as described by Soil Survey Staff (2006). Soil porosity of the sampled soil showed a range of 32 – 42 % and fall within the range (30 – 60%) that enhance better aeration, water movement and retention, and microbial activities as reported by Oti, (2007). All other measured water quality parameters for the irrigation water except pH which ranged (5.31-5.40) fell within the permissible limits for irrigation as stipulated by Food and Agricultural Organization (FAO).

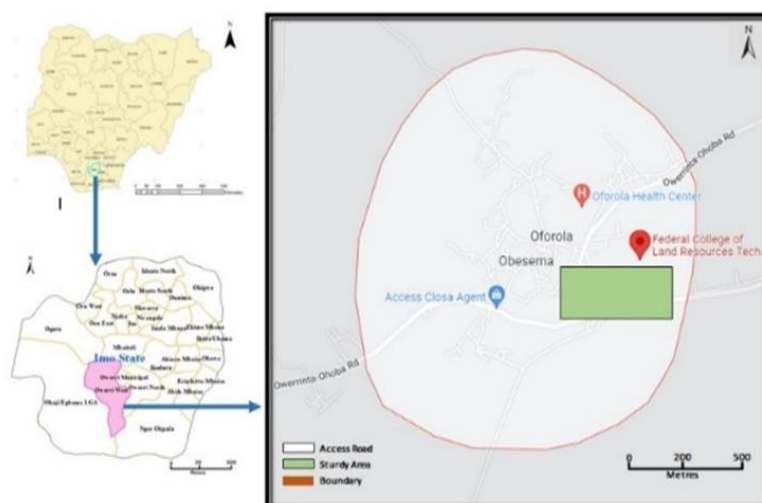


Figure 1: Map of Imo State Showing the Study Area

Table 1: Average Daily Climatological Data (2013-2023) for the Months of the Experiment

Month	Tmin (°C)	Tmax (°C)	RH (%)	Wind (km/day)	Speed	Net (MJ/m ² /day)	Radiation	Sunshine Hours	ETc (mm/day)
Sept.	22.6	30.6	85	137		16.3		4.5	3.51
Oct.	22.4	32.0	82	113		16.6		5.0	3.83
Nov.	17.2	25.9	74	93		16.8		5.6	3.19
Dec.	19.8	34.1	63	103		18.7		7.3	4.23
Jan.	20.3	34.3	69	118		19.7		7.7	5.00
Feb.	21.7	30.9	70	118		20.1		7.3	4.40
Mar.	23.2	33.4	80	120		19.3		6.8	4.46

Source: Igbojionu, (2024)

Methods of Data Generation

Estimation of Reference Evapotranspiration (ET_o)

The reference evapotranspiration (ET_o) was estimated by the use of CROPWAT 8.0 software based on equation 1. CROPWAT 8.0 is a windows computer program for calculation of crop water requirements based on soil, climate and crop data. In addition, the program permits the development of irrigation schedules for different management conditions and calculation of scheme water supply for varying

crop patterns. The weather data presented in Table 1 for maximum and minimum temperature, sunshine hours, maximum and minimum relative humidity, solar radiation and wind speed for ten years obtained from the internet-based weather station of the Federal College of Land Resources Technology, Owerri were used.

$$ET_o = \frac{0.408 \Delta(R_n - G) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

Where,

ET_o = reference evapotranspiration [mm day⁻¹],
 R_n = net radiation at the crop surface [MJ m⁻² day⁻¹],
 G = soil heat flux density [MJ m⁻² day⁻¹],
 T = mean daily air temperature at 2 m height [°C],
 u₂ = wind speed at 2 m height [m s⁻¹],
 e_s = saturation vapor pressure [kPa],
 e_a = actual vapor pressure [kPa],
 e_s – e_a = saturation vapor pressure deficit [kPa],
 Δ = slope of the vapor pressure curve [kPa °C⁻¹],
 γ = psychrometric constant [kPa °C⁻¹].

Estimation of Crop Water Requirement of Cucumber

The water requirement of cucumber was calculated using equation 2 proposed by Allen *et al.*, (1998) and applied by Al-Kalifa *et al.*, (2013) and the K_c values presented in Table 2. The gross water requirement was then calculated according to Choudhary and Kadam (2006) as crop water requirement (ET_c) divided by system efficiency (90%) and presented in Tables 3

$$ET_c = ET_o \times K_c \quad (2)$$

Where:

ET_c = crop evapotranspiration (mm/day)

ET_o = reference evapotranspiration (mm/day) K_c = crop coefficient (dimensionless)

T₁ (100%) received 186mm, T₂ (75%) received 139.5mm, T₃ (50%) received 93mm and T₄ (25%) received 46.5mm.

Calculation of Irrigation Treatments Used to Develop Irrigation Schedule

For each crop growth stage of cucumber, the number of irrigations, depth of water to apply, time taken to apply depth of water per irrigation were calculated according to Singh (2012) as follows and displayed in Table 3:

$$\text{Number of irrigations} = \frac{\text{Length of growth stage}}{\text{Irrigation interval}} \quad (4)$$

$$\text{Depth of water to be applied per irrigation} = \frac{\text{Irrigation treatment}}{\text{Number of irrigations}} \quad (5)$$

$$\text{Precipitation of the emitter} = \frac{\text{Emitter discharge}}{\text{Area occupied by an emitter}} \quad (5)$$

$$\text{Time taken to apply depth of water per irrigation} = \frac{\text{Depth of water per irrigation}}{\text{Emitter precipitation}} \quad (6)$$

Experimental Design and Treatments

The Cucumber (*Cucumis sativus*) (African Giant variety) were planted three per hole in seedbeds at an intra-row and

inter-row spacing of 33 cm x 50 cm, and 1 m between seedbeds as buffer in a greenhouse facility. After sprouting the plants were thinned to one per hole was uniformly watered within the first two weeks to establish them (Zakka *et al.*, 2020). Thereafter, the irrigation treatments for each growth stage presented in Tables 4-7 were imposed simultaneously under a drip irrigation system operated in the morning throughout the cropping seasons of September – November, 2023.

A randomized complete block design (RCBD) of 4 by 3 factorial of 12 treatments with three replications, was used in the study. The experimental treatment consists of two factors which are irrigation level (W) of crop evapotranspiration (ET_c) and irrigation interval (D), days of irrigation intervals. The irrigation level W, consists of four levels of W₁, W₂, W₃, W₄ for 100% of ET_c, 75% of ET_c, 50% of ET_c and 25 % of ET_c respectively. While the irrigation interval D, of 1 day, 5 days and 10 days are D₁, D₅ and D₁₀ respectively.

Cucumber Yield Data Collection and Analysis

Random sampling technique was used to select and mark cucumber plants in the plots crop yield parameters such as number of fruits per plant, weight of fruit, size, and marketable yield were collected and recorded during harvest. The number of fruits per plot, weight of fruit, length and diameter were determined by counting, weighing balance and digital vernier caliper respectively. The total number, average weight, and size of fruit per treatment was determined at the end of the late season. The cucumber yield collected for yield parameters were subjected to statistical analysis using Analysis of Variance (ANOVA) by means SARS software analytical technique to evaluate the effect of different irrigation treatments on the cucumber yield at (p<0.05).

RESULTS AND DISCUSSION

Results

Computed Values for Different Growth Stages of Cucumber

Table 2 presents the actual values of crop coefficients (K_c) for each growth stage of the cucumber plant. The K_c value is a multiplier used to calculate how much water the cucumber plant needs for each growth stage.

Table 2: Computed K_c Values for Different Growth Stages of Cucumber

Growth Stage	Length of Growth Stage (days)	Date	K _c Value
Initial	10	21st Sept - 30th Sept	0.45
Crop development	20	1st Oct. - 20th Oct.	0.80
Mid-season	30	21st Oct. - 19th Nov.	0.80/0.82
Late season	10	20th Nov. - 29th Nov.	0.82

The actual K_c values for October and November (overlapping months) were computed as 0.80 and 0.82 respectively using weighted approach. The K_c value changes as the cucumber plant grows. The total growth duration for the cucumber variety used in this study is 90 days comprising initial stage, crop development, mid-season and late season of 10, 20, 30 and 10 days respectively. During initial stage of growth (10 days) young cucumber plants had small leaves (K_c=0.45) and lost less water to evaporation, while during crop development (20 days), the plant leaves grew rapidly with increase in water

use (K_c= 0.80). The mid-season is the longest stage (30 days) when cucumber plants developed full canopy cover, produced fruits with peak water demand depicted by K_c value of 0.82. The late season (10 days) is the last growth stage of the cucumber plant when growth and water demand slowed down as the plant matured and dried up

Crop Water Requirement (ET_c) of Cucumber Crop

Table 3 displays the actual daily water and total water used by the cucumber crop in each growth stage.

Table 3: Growth Stages and Crop Water Requirement (ETc) for Cucumber

Stage of Growth	Length of Growth Stage (days)	Daily ETc (mm)	Total for Period (mm)	Total Growth Stage ETc (mm)
Initial	10 (Sept.)	1.750	17.55	17.55
Crop Development	20 (Oct.)	3.3867	67.734	67.73
Mid-Season	30	Varies	92.4753	92.46
- October (11 days)	11	3.3867	37.2537	
- November (19 days)	19	2.9064	55.2216	
Late Season	10 (Nov.)	2.9064	29.064	29.06

Daily crop water requirement (ETc) was calculated as a product of crop coefficient (Kc) and reference evapotranspiration (ETo). Kc values from Table 3 were used while ETo was estimated by the use of CROPWAT 8.0 software based on weather data presented in Table 1. Total ETc for each growth stage is the product of daily ETc (mm) and length of stage (days). Total water requirement for the entire growing season is the sum of all stages: 17.55 + 67.73 + 92.46 + 29.06 = 207 mm.

Developed Irrigation Schedule for Initial Growth Stage of Cucumber

Table 4 details the irrigation parameters for the initial growth stage of cucumber seedlings and reveals a clear inverse relationship between irrigation frequency and application depth.

Table 4: Irrigation Schedule for Initial Growth Stage of 10 Days for Cucumber Production

Treatment	Treatment Combination	Irrigation Interval (days)	No. of Irrigations	Depth of Water per Irrigation (mm)	Time Taken
W1: 100% ETc = 18 mm	W1D1	1	10	1.8	17 min
	W1D5	5	2	9.0	1h 23 min
	W1D10	10	1	18.0	2h 46 min
W2: 75% ETc = 13.5 mm	W2D1	1	10	1.35	11 min
	W2D5	5	2	6.75	1h 2 min
	W2D10	10	1	13.5	2h 5 min
W3: 50% ETc = 9 mm	W3D1	1	10	0.9	8 min
	W3D5	5	2	4.5	42 min
	W3D10	10	1	9.0	1h 23 min
W4: 25% ETc = 4.5 mm	W4D1	1	10	0.4	4 min
	W4D5	5	2	2.25	21 min
	W4D10	10	1	4.5	42 min

Under a fixed 10-day water volume of 18 mm (for the 100% ETc treatment), the data demonstrates that a daily irrigation schedule (D1) involved frequent but shallow applications, which resulted in a lower daily time commitment. In contrast, extended intervals such as D5 and D10 required fewer irrigation events, but each event must be deeper and more time-intensive to deliver the required water. Furthermore, it shows that deficit irrigation treatments (W2-W4) systematically reduced the total water applied, which consequently decreased the application depth and time required per event. Cucumber growers, the schedule reveals are faced with the challenge between daily watering which requires more labour but promotes consistent moisture, and

less frequent watering, which encourages deeper soil moisture penetration crucial for establishing strong seedlings.

Developed Irrigation Schedule for Crop Development Stage of Cucumber

Table 5 outlines the trade-offs between irrigation frequency and application depth for the 20-day cucumber development stage, showing that while the total water volume (68 mm for 100% ETc) remains constant per treatment, applying it less frequently (every 10 days) requires deeper, longer irrigation events per session, whereas daily irrigation requires more frequent but much shorter applications, with deficit irrigation treatments (W2-W4) proportionally reducing the water depth and time required per event.

Table 5: Irrigation Schedule for Crop Development Stage of 20 Days for Cucumber Production

Treatment	Treatment Combination	Irrigation Interval (days)	No. of Irrigations	Depth of Water per Irrigation (mm)	Time Taken Per Irrigation
W1: 100% ETc = 68 mm	W1D1	1	20	3.4	31 min
	W1D5	5	4	17.0	2h 37 min
	W1D10	10	2	34.0	5h 14 min
W2: 75% ETc = 51 mm	W2D1	1	20	2.55	23 min
	W2D5	5	4	12.75	1h 58 min
	W2D10	10	2	25.5	3h 55 min
W3: 50% ETc = 34 mm	W3D1	1	20	1.7	16 min
	W3D5	5	4	8.5	1h 19 min
	W3D10	10	2	17.0	2h 37 min

Treatment	Treatment Combination	Irrigation Interval (days)	No. of Irrigations	Depth of Water per Irrigation (mm)	Time Taken Per Irrigation
W4: 25% ET _c = 17 mm	W4D1	1	20	0.85	8 min
	W4D5	5	4	4.25	39 min
	W4D10	10	2	8.5	1h 19 min

Developed Irrigation Schedule for Mid-Season Stage of Cucumber

Table 6 details the irrigation strategies for the water-sensitive mid-season stage of cucumber production, demonstrating that

for a constant total water volume applied over 30 days (92.46 mm for the full requirement, W1), the chosen irrigation frequency dictates the depth and duration of each watering event.

Table 6: Irrigation Schedule for Mid-Season Stage of 30 Days for Cucumber Production

Treatment	Treatment Combination	Irrigation Interval (days)	No. of Irrigations	Depth of Water per Irrigation (mm)	Time Taken Per Irrigation
W1: 100% ET _c = 92.46 mm	W1D1	1	30	3.08	29 min
	W1D5	5	6	15.41	2hr 22 min
	W1D10	10	3	30.82	4hr 45 min
W2: 75% ET _c = 69.345 mm	W2D1	1	30	2.31	21 min
	W2D5	5	6	11.56	1hr 55 min
	W2D10	10	3	23.12	3hr 33 min
W3: 50% ET _c = 46.23 mm	W3D1	1	30	1.54	14 min
	W3D5	5	6	7.71	1hr 11 min
	W3D10	10	3	15.41	2hr 22 min
W4: 25% ET _c = 23.12 mm	W4D1	1	30	0.77	7 min
	W4D5	5	6	3.85	36 min
	W4D10	10	3	7.71	1hr 11 min

Daily irrigation (D1) requires 30 brief, shallow applications to maintain consistent soil moisture, while extended intervals (D5, D10) consolidate the water into fewer but significantly deeper and more time-intensive applications, with deficit irrigation treatments (W2-W4) proportionally reducing the total water volume, application depth, and time required per event. This highlights a clear trade-off between irrigation

frequency, labor time, and potential water savings against the risk of plant stress during this critical growth phase.

Developed Irrigation Schedule for Late Season Stage of Cucumber

Table 7 outlines the irrigation plan for the late-season stage of cucumber production. It shows a fixed total water requirement of 29.06 mm over 10 days for the full treatment (W1).

Table 7: Irrigation Schedule for Late-Season Stage of 10 Days for Cucumber Production

Treatment	Treatment Combination	Irrigation Interval (days)	No. of Irrigations	Depth of Water per Irrigation (mm)	Time Taken Per Irrigation
W1: 100% ET _c = 29.06 mm	W1D1	1	10	2.91	27 min
	W1D5	5	2	14.53	2hr 17 min
	W1D10	10	1	29.06	4hr 29 min
W2: 75% ET _c = 21.795 mm	W2D1	1	10	2.18	20 min
	W2D5	5	2	10.90	1hr 41 min
	W2D10	10	1	21.80	3hr 21 min
W3: 50% ET _c = 14.9 mm	W3D1	1	10	1.49	14 min
	W3D5	5	2	7.45	1hr 9 min
	W3D10	10	1	14.90	2hr 18 min
W4: 25% ET _c = 7.265 mm	W4D1	1	10	0.727	7 min
	W4D5	5	2	3.63	34 min
	W4D10	10	1	7.27	1hr 7 min

The chosen irrigation frequency creates a direct trade-off. Daily irrigation (D1) requires frequent but shallow applications. This means 10 irrigations of 2.91 mm, each

taking 27 minutes. This strategy maintains consistent moisture.

In contrast, a single irrigation (D10) applies the entire requirement at once. This is one deep application of 29.06 mm, taking 4 hours and 29 minutes. Deficit irrigation strategies (W2-W4) systematically use less total water. This consequently lowers the depth and time needed for each application. This offers farmers a critical choice. They can choose frequent watering for higher labor but potentially better yield and fruit quality. Alternatively, they can choose

less frequent irrigation to save water and labor, accepting some plant stress as the harvest ends.

Effect of Irrigation Regime and Interval on Cucumber Yield Parameters

Table 8 demonstrates that both irrigation amount (IA) and application interval (II) significantly impact cucumber yield and fruit quality. This impact is both statistically significant ($p < 0.05$) and substantial.

Table 8: Effect of Irrigation Regime and Interval on Cucumber Yield Parameters

Treatment	Yield (ton/ha)	Ave. Fruit Weight (g)	Ave. Fruit Length (cm)	Ave. Fruit Diameter (cm)	No. of Fruits per Plant
W1D1	121.00	0.565	35.5	5.95	13
W1D5	89.17	0.557	34.6	5.85	10
W1D10	72.33	0.510	33.7	5.75	9
W2D1	73.50	0.531	35.5	5.71	9
W2D5	63	0.478	34.6	5.69	9
W2D10	50.17	0.422	31.6	5.30	8
W3D1	47.17	0.356	32.7	5.62	8
W3D5	31.17	0.328	32.1	5.54	7
W3D10	29.00	0.272	31.6	5.41	7
W4D1	19.83	0.213	31.3	5.31	6
W4D5	17.17	0.210	30.6	5.23	5
W4D10	10.83	0.181	29.4	5.13	4
IA	*	*	*	*	*
II	*	*	*	*	*
IA:II	*	*	*	*	*

IA is irrigation amount, II is irrigation interval, IA: II is interaction between irrigation amount and irrigation interval, * means significant ($p < 0.05$)

The best results were achieved by combining the full water requirement (W1) with daily irrigation (D1). This treatment yielded 121.00 tons per hectare. It also produced the largest fruits (0.565 g, 35.5 cm, 5.95 cm) and the most fruits per plant (13).

Reducing the irrigation amount (from W1 to W4) caused severe, progressive declines in all yield parameters. Similarly, extending the irrigation interval (from D1 to D10) also caused major declines at every water level. For instance, even with full water (W1), extending the interval from daily (D1) to 10 days (D10) reduced yield by 40%. This is a drop from 121 to 72.33 tons per hectare. It also reduced average fruit weight by 8%, average fruit length by 5%, average fruit diameter by 3.4% and fruit number per plant by 31% representing drops of 0.565-0.510 g, 35.5-33.7 cm, 5.95-5.75 cm and 13-9 respectively.

A critical interaction was found: the negative effect of infrequent watering is strongest under full irrigation. However, it remains harmful across all water levels. This underscores that frequent irrigation is essential as it maintains soil moisture and protects fruit development from drought impacts.

Discussion

Integrating scheduling data (Tables 4–7) with yield outcomes (Table 8) reveals that irrigation amount (IA) ($p < 0.01$), irrigation interval (II) ($p < 0.05$), and their interaction (IA $\times$ II) ($p < 0.05$) significantly govern yield, fruit size, and fruit number. The findings highlight critical interactions between IA, II, and crop phenological stage, offering actionable strategies for sustainable production. Irrigation amount (IA), defined as a percentage of crop evapotranspiration (ETc), was the primary yield determinant. Full irrigation (100% ETc, W1) maximized performance, achieving peak yield (121

ton/ha), fruit weight (0.565 g), fruit length (35.5 cm), and fruit number (13/plant). Conversely, severe deficit (25% ETc, W4) caused drastic declines: 91% lower yield (10.83 ton/ha), 68% reduced fruit weight, 18% shorter fruits, 14% smaller fruits and 69% fewer fruits. This aligns with water deficit impairing photosynthesis, cell expansion, and nutrient assimilation (Kirda, 2002; Wang et al., 2023).

The weight, fruit size and fruit number per plant varied significantly between treatments, according to the results (Table 8). Higher fruit weights, longer fruits, larger fruits and higher numbers per plants were associated with full irrigation. In general, full irrigation increased fruit weight, which is consistent with research demonstrating that growth parameters are enhanced by an adequate water supply (Rahil, 2015). The highest cucumber yield of 121 t/ha was attained with 100 % ETc daily irrigation which was 39%, 61% and 84% higher than that obtained with 75% ETc, 50% ETc and 25% ETc applied daily respectively while the lowest yield of 19.83 t/ha was recorded with 25% ETc. This result conforms with the findings of Al-Omran and Lucki (2012) who reported yields as much as between 91– 150 t/ha for the indeterminate cucumber cultivar in a greenhouse. Furthermore, Sahin et al., (2015) and Ayas and Demirtas (2009) obtained highest yield at 100 % of ETc in a greenhouse. Recent work by Kareem et al. (2023) strongly supports this, demonstrating that applying 100% of crop evapotranspiration (ETc) resulted in a significantly higher marketable yield (116.3 t/ha) compared to 50% ETc (37.8 t/ha) for greenhouse cucumbers. In another study, Lawal et al. (2020) achieved the highest average maize yield (3.58 kg/ha) with 100 % ETc irrigated every 7 days while the least (1.2 kg/ha) was with 50 % ETc irrigated every 13 days, further corroborating the results of this study. Similarly, Zakka et al. (2020) in their study on cucumber in two planting seasons, reported the highest yields (21 t/ha and 20.3 t/ha) from the treatment with 100 % ETc, irrigated daily while the lowest (15.5 t/ha and 16.5 t/ha) were obtained from the treatment with 60 %, irrigated every 4 days. Our findings

further align with those of Peter et al. (2024) who also achieved the highest yield of 14.4 and 14.42 t/ha with daily full irrigation and 4.133 and 4.822 ton/ha with deficit irrigation at 7-9 weeks for marketer and pointset cucumber cultivars respectively

However, Mamun-Hossain et al., (2018) reported 85 % of ETc as the treatment that produced the highest fruit yield compared to 100 % of ETc in a greenhouse. The total yield from all treatments was 624.34 t/ha. The yields for each treatment were compared and found to be statistically significant ($p > 0.05$)

With respect to average fruit weight, full irrigation amount applied daily produced the highest average fruit weight of 0.565 g while the lowest average fruit weight of 0.181 g was obtained with 25% irrigation amount applied every ten days representing a reduction in average fruit weight of 68%. This reduction agrees with the findings of Musa, (2018), Parkash et al. (2021), and Zakka et al. (2020) that as irrigation amount decreased, the yield of cucumber decreased indicating that irrigating a plant below full irrigation amount has the tendency to reduce yield. This also conforms with the findings of Sahin et al. (2015); Mamun-Hossain et al., (2018) who reported the highest average fruit weight at 100 % ETc and the lowest value at 40% ETc. Similarly, Al-Khalifa et al., (2013) obtained the best yield and bunch weight of banana at 100 % of ETc, irrigated daily. Goenaga and Irizarry, (1998) the use of Class A pan factors of 0.0-1.0 showed significant improvement in yield and yield components of banana and attributed this result to increase in water applied. The results of statistical analysis showed that irrigation amount had a significant ($p < 0.05$) effect on fruit weight.

The highest average fruit length (35.5 cm) was produced with 100% ETc irrigated daily while the lowest average fruit length (29.4 cm) was obtained with 25% ETc irrigated every 10 days. These results agree with Sahin et al. (2015) who obtained the highest average length of cucumber of 15.95 cm from 1.0 of cumulative evapotranspiration from Class A pan. Mamun-Hossain et al. (2018) and Omotade and Babalola (2019) reported the highest average fruit length of 18.33 cm at 80 % and 26.55 cm at 65 % respectively. Peter et al. (2024) in a similar study on two cucumber cultivars obtained the longest cucumber fruits (15.52 cm and 16.04 cm) with daily full irrigation and the shortest fruits (10.66 cm and 10.98 cm) with irrigation at 7-9 weeks. Their results are, however, much lower than that achieved in this study. Our findings show that moisture stress during the reproductive stage has a detrimental effect on fruit size reducing both diameter and length. This emphasizes the need for optimal soil moisture levels to support photosynthesis and fruit development. These findings align with those of Alomran and Louki (2011), who also highlighted the importance of optimal soil moisture levels for achieving high yields and quality in cucumbers. Results of statistical analysis indicated that irrigation level significantly affected average length fruit at $p < 0.05$.

Fruit diameter significantly responded to deficit irrigation. The decrease in the amount of water applied resulted in a corresponding decrease in fruit diameter. In this study, the highest average fruit diameter (5.95 cm) was achieved with 100% ETc irrigated daily while the lowest value (5.31 cm) was recorded at 25 % ETc irrigated every 10 days. This result aligns with Celiktopuz et al. (2025) who stated that plants under full irrigation treatments exhibited fruit diameter 4.9% superiority compared to the plants receiving 50% deficit irrigation. This further resonates with Omotade and Babalola (2019) who reported the highest average fruit diameter of 5.87 cm with (80%) while the lowest average fruit diameter was obtained with 40 %. In contrast, Adeosun et al. (2025) in a

similar study on cucumber greenhouse reported larger fruit diameter (44.50 cm) at deficit irrigation 70% compared to full irrigation (43.08 cm). The results of statistical analysis showed that irrigation amount had a significant effect on average fruit diameter of cucumber. Generally, frequent daily irrigation leads to longer and larger cucumber fruits due to consistent soil moisture and cell turgor while longer intervals can reduce fruit size by causing water stress that limits cell expansion

The average number of fruits per plant varied from 13– 6 representing a reduction of 53.9%. The highest average number of fruits per plant (13) was recorded from 100% ETc irrigated daily while the least number (6) was obtained from 25% ETc irrigated every 10 days. This finding is consistent with Al-Khalifa (2013) who asserted that number of hands per bunch and number of fingers per bunch of bananas were linearly related to the quantity of water to the crop. It is also consistent with the findings of Peter et al. (2024) who found the highest number of fruits per plant of 13.67 and 14.83 at full irrigation, and the least number of 7.42 and 7.17 with deficit irrigation for marketer and pointset cucumber cultivars respectively. The result of the statistical analysis showed that there were significant differences in the number of fruits per plant, demonstrating that deficit irrigating significantly affects number of fruits per plant.

CONCLUSION

Cucumber yield and quality were highly responsive to both irrigation volume (IA) and timing (II). While full irrigation (100% ETc) with daily applications maximizes productivity, adopting 75% ETc with daily irrigation offers a sustainable compromise, reducing water use by 25% with moderate yield loss. Infrequent irrigation (D5/D10) exacerbates water stress even under high IA, highlighting the need for precision scheduling.

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